ANSWER KEY

Class XII – Computer Science – Unit Test – 1

SECTION A (17 × 1 = 17 Marks)

1. True
2. c) List
3. a) True
4. c) func
5. c) ming
6. 2 5 8 11 14
7. 5
8. c) is
9. c) Not Found
10. a) 15@25@
11. a) 10 -- 15##15
12. a) ['Hello', 'World', 'Python']
13. c) from math import sqrt, ceil
14. a) Both A and R are True and R is the correct explanation
15. b) input()
16. True
17. a) True

SECTION B (5 × 2 = 10 Marks)

18A. Difference between break and continue

break – exits the loop entirely.
continue – skips the current iteration and continues.

```
for i in range(5):
    if i == 3:
        break
    print(i)
```

```
for i in range(5):
    if i == 3:
        continue
    print(i)
```

18B. Explicit vs Implicit Type Conversion

Implicit: Python converts automatically.

```
x = 5 + 3.2
```

Explicit: Programmer converts manually.

```
x = int(3.8)
```

19. Corrected Code

```
def modify_string(s):
    if len(s) < 2:
        return s
    new_str = s[1:-1]
    return new_str

result = modify_string("Python")
print("Modified string:", result)
```

20A.

I. `"hello world".upper()`

II. `sentence.index("is")`

20B. Output

```
('Learn Python programming ', 'with', ' fun')
2
```

21A. remove_item()

```
def remove_item(L, n):
    if n in L:
        L.remove(n)
    else:
        print("Element not found")
```

21B. add_student()

```
def add_student(students, roll_no, name):
    if roll_no in students:
        print("Student already exists")
    else:
        students[roll_no] = name
```

22. Differences

I. Local vs Global Variable

- **Local:** Declared inside a function; accessible only there.

- **Global:** Declared outside all functions; accessible everywhere.

II. list() vs tuple()

- **list():** Mutable, uses []
- **tuple():** Immutable, uses ()

SECTION C (2 × 3 = 6 Marks)

23. Count occurrences of "the"

```
def count_the():
    count = 0
    with open("story.txt", "r") as f:
        for word in f.read().split():
            if word.lower() == "the":
                count += 1
    print("Count:", count)
```

24A. Output

```
ODE234*
```

24B. Output

```
['DELHI', 'BANGALORE', 'CHENNAI']
```

SECTION D (3 × 4 = 12 Marks)

25. List Operations

```
L = [10, 20, 30, 20, 40, 20]
print("Max:", max(L))
print("Min:", min(L))
print("Sum:", sum(L))

n = int(input("Enter number to search: "))
count = 0
for x in L:
    if x == n:
        count += 1
print("Frequency of", n, "is", count)
```

26A. Employee Dictionary

```
emp = {"A":60000, "B":45000, "C":52000, "D":30000, "E":75000}

for k, v in emp.items():
    if v > 50000:
        print(k, v)

print("Total salary:", sum(emp.values()))
```

26B. List – max, min, sum, sorted

```
L = []
for i in range(10):
    L.append(int(input()))

print("Max:", max(L))
print("Min:", min(L))
print("Sum:", sum(L))
print("Sorted:", sorted(L))
```

27. Outputs

Code 1: 6

Code 2: (20, 30, 40)

Code 3: dict_keys(['a', 'b', 'c', 'd'])

Code 4: 2 4 8

SECTION E (1 × 5 = 5 Marks)

28A. Sum of Even Numbers

```
def sum_even(L):
    s = 0
    for x in L:
        if x % 2 == 0:
            s += x
    return s

nums = []
for i in range(5):
    nums.append(int(input()))

print("Sum of even numbers:", sum_even(nums))
```

28B. Count Vowels & Consonants

```
s = input("Enter a string: ")
vowels = "aeiouAEIOU"
v = c = 0

for ch in s:
    if ch.isalpha():
        if ch in vowels:
            v += 1
        else:
            c += 1

print("Vowels:", v, "Consonants:", c)
```